

BIOMECHANICS (EPHE 341)

Linear Kinetics

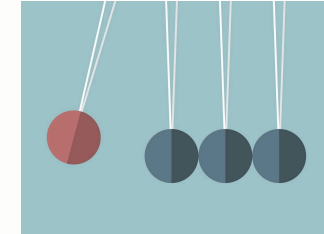
Optional reading: McGinnis Chpt 3

Objectives

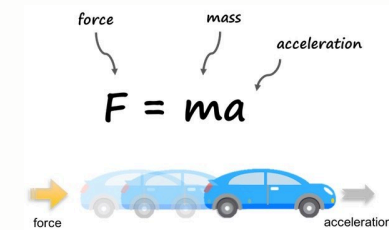
To learn/review;

- Newton's 3 laws
- Kinetics equations and relationships
- Force
- Momentum
- Impulse
- Coefficient of restitution

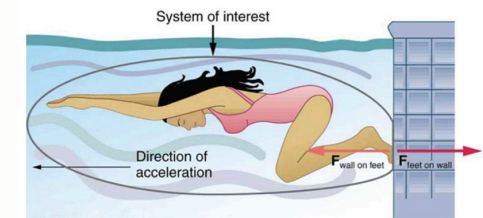
1st Law – Inertia



2nd Law – Acceleration

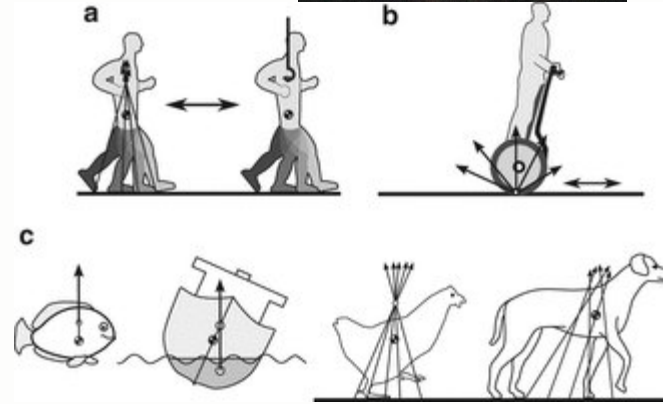
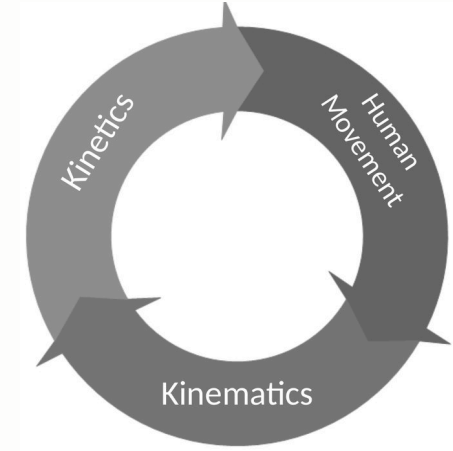
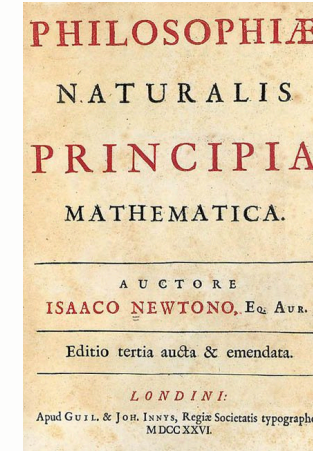
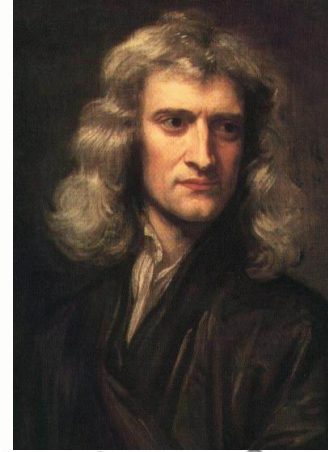


3rd Law – Action / Reaction



Kinetics:

The concepts of mass, force, and energy as they affect motion.



Newton's 1st Law : Law of Inertia

An object at rest tends to stay at rest and an object in uniform motion tends to stay in uniform motion unless acted upon by a **net external force**.

- uniform = unchanging = consistent
- Net = total after all forces are added together

What is Inertia?

- derived from the latin for "idleness" or "laziness"
- resistance to a change in state of motion

Mass quantifies inertia. Inertia is not a separate property you measure on its own — the kilogram *is* the measure of it. Double the mass and you have



More mass, more inertia — harder to start, harder to stop, harder to turn.

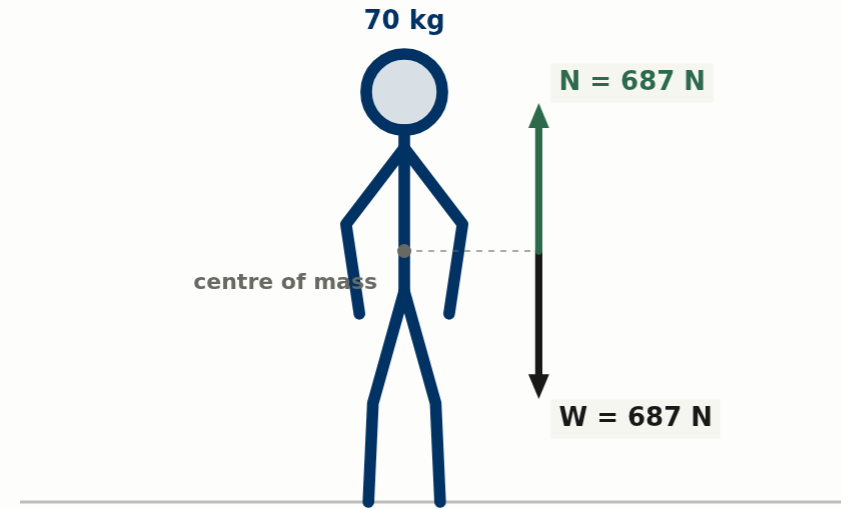
Static equilibrium

An object is at rest when the net forces acting on it sum to zero.

- There is no motion
- Gravity is acting downward
- The ground contact force is acting up
- The net vertical force is zero

$$\Sigma F_z = 0 \quad \Sigma F_x = 0 \quad \Sigma F_y = 0$$

This individual is in static equilibrium. **Is there such a thing as dynamic equilibrium?**



static equilibrium

$$\checkmark \Sigma F_x = 0 \text{ N}$$

$$\checkmark \Sigma F_y = 0 \text{ N}$$

$$\checkmark \Sigma F_z = 0 \text{ N}$$

$$a = \Sigma F / m = 0.00 \text{ m/s}^2$$

static equilibrium — at rest, and staying that way

Equilibrium is not one equation, it is **three**: the forces have to sum to zero along every axis independently. Standing still there are only two forces in play — the weight the Earth pulls down with, and the contact force the ground pushes up with — and because they are equal and opposite, all three sums come out zero at once. Add a push along any one axis and that sum breaks on its own, and the body accelerates in that direction and no other.

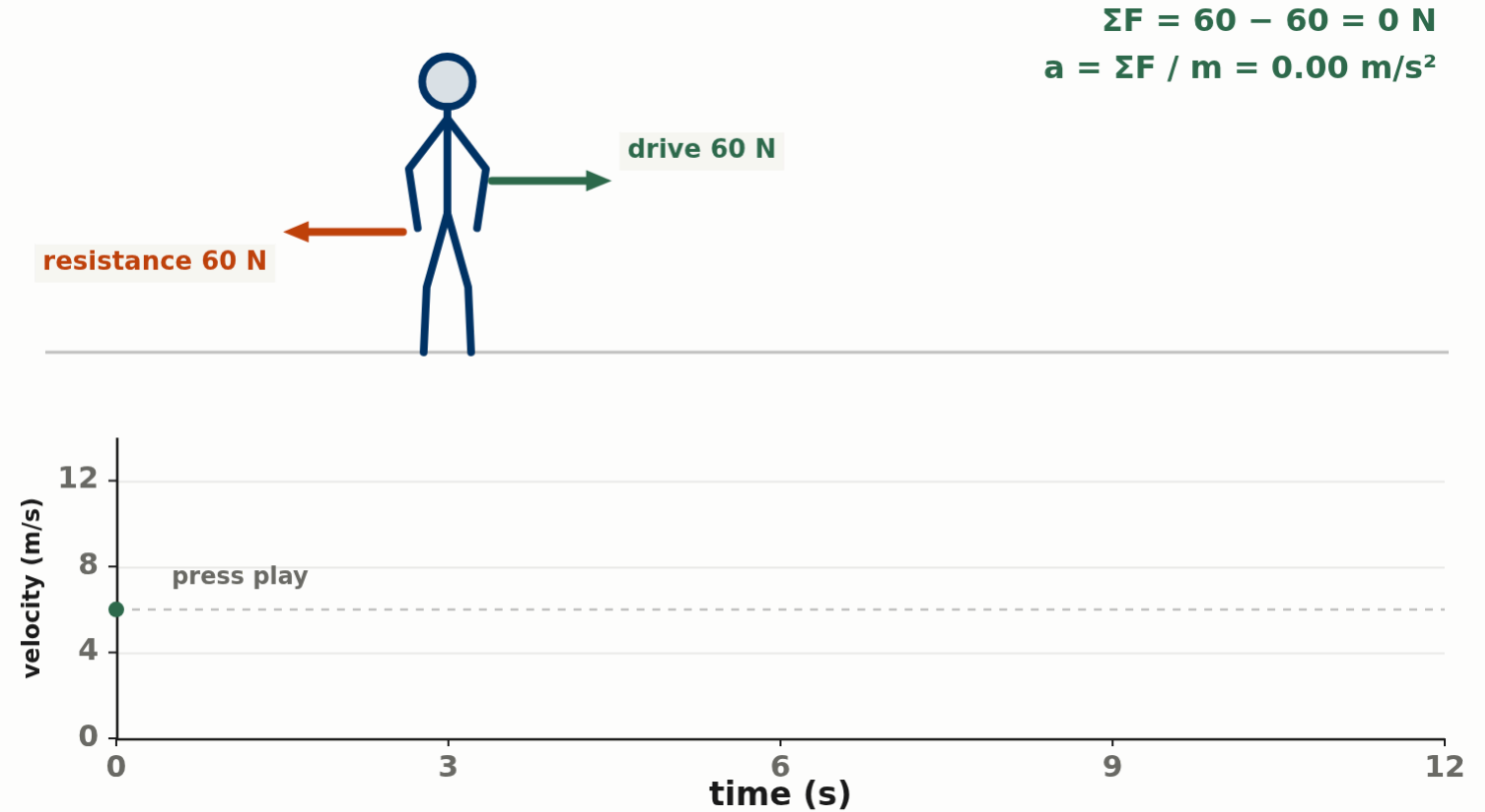
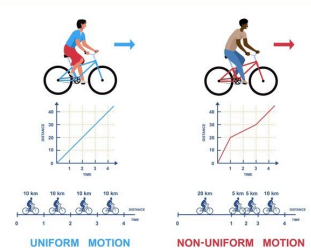
Push
along x  0 N

Push
along y  0 N

Dynamic equilibrium

When is an object in uniform motion? This is **Dynamic Equilibrium**.

- Uniform motion means constant velocity.
- An object will maintain its velocity unless accelerated or decelerated.
- Net forces are equal to zero but an object is in motion



dynamic equilibrium — moving, and staying that way · $v = 6.00 \text{ m/s}$

Uniform motion means **constant velocity**, and constant velocity means zero acceleration, which by $F = ma$ means zero net force. So the force diagram here is the same one as for a body standing still — equal and opposite, summing to nothing. Motion does not need a force to keep it going; only a **change** of motion does. Nudge the drive away from the resistance and watch the trace tilt.

Drive
force

60 N

Resistance

60 N

Momentum

The momentum of an object can be thought of as the tendency for an object to resist change in motion and continue to move in its direction of travel.



The football player's momentum will resist change in the state of motion.
He will be difficult to decelerate



The cyclist's momentum will resist change in the state of motion. He will
keep moving forward

Momentum

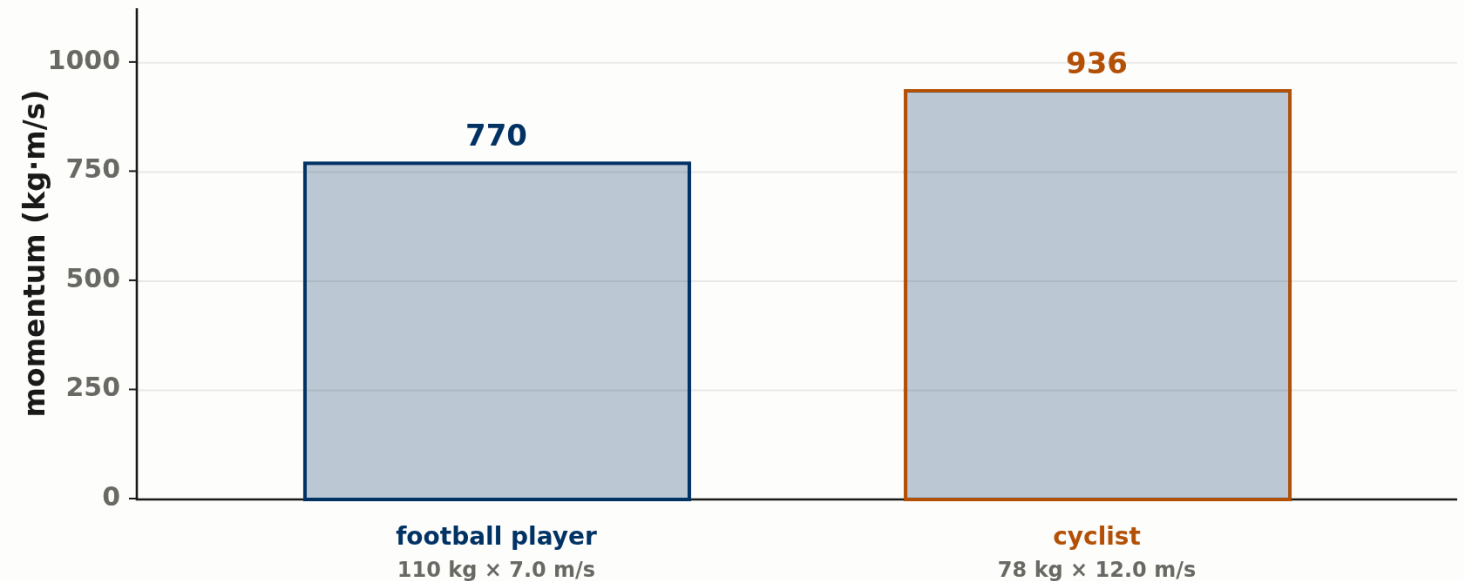
Momentum is related to inertia but not the same. Inertia is an object's tendency to resist change in the state of motion (at motion and rest). However during motion an object has momentum.

That is, an object resists its change in state of motion and this resistance is related to its **mass** (inertia) and **velocity**.

$$P = mv$$

where P is the object's linear momentum, m is the object's mass and v is the object's velocity. SI units: kg·m/s

Remember that velocity is a vector quantity, so momentum will have a magnitude and direction.



$P = mv$ · 770 against 936 kg·m/s

Momentum is mass **and** velocity. Make the lighter one fast enough and it carries as much as the heavier one — try it. Velocity is a vector, so momentum has a direction too.

Player mass 110 kg
Cyclist mass 78 kg

Player speed 7.0 m/s
Cyclist speed 12.0 m/s

Momentum

Understanding Momentum is really important when dealing with things that collide.

- Baseball
- Soccer
- Football players

The outcome of collisions during these sports can be explained by quantifying the momentum of the system.



Law of Conservation of Momentum

In the absence of external forces, the total momentum of a given system is conserved.

$$P_{\text{before}} = P_{\text{after}} \quad \text{or} \quad (mv)_{\text{before}} = (mv)_{\text{after}}$$

The law of conservation of momentum helps us understand collisions — the sum of the momentum before the collision must equal the sum of the momentum after the collision.

Collisions range between elastic and inelastic collisions:

- **Elastic collisions** — objects rebound off of each other
- **Inelastic collisions** — objects stick together and usually become distorted and generate heat

Conserved means the total is the same number before and after — not that each object keeps what it had. The collision *redistributes* momentum; it does not create or destroy any.

$$m_A v_{Ai} + m_B v_{Bi} = m_A v_{Af} + m_B v_{Bf}$$

Coefficient of restitution

Most collisions are neither perfectly elastic nor perfectly inelastic but somewhere in between.

The **coefficient of restitution** is a means of quantifying how elastic the collisions are.



Coefficient of restitution

The coefficient of restitution is the absolute value of the ratio of the **velocity of separation** to the **velocity of approach**.

The **velocity of approach** is the difference between the two velocities just *before* the collision — how fast they are closing.

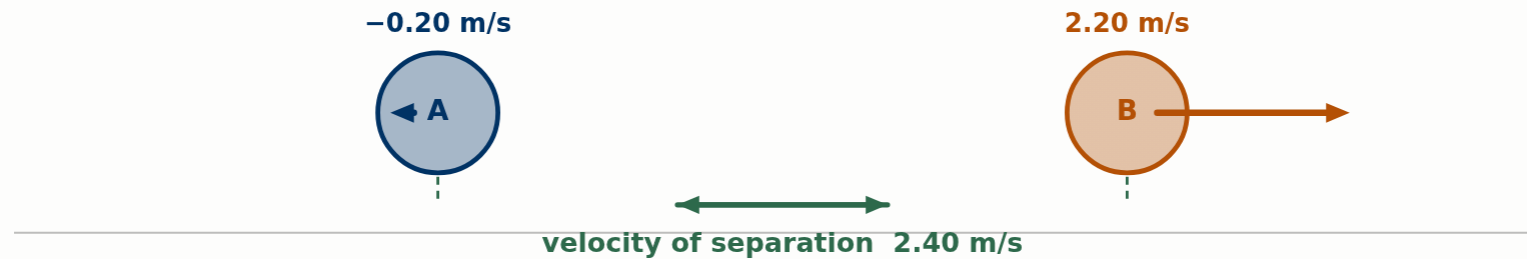
The **velocity of separation** is the difference just *after* it — how fast they are moving apart.

Before the collision



$$e = \text{velocity of separation} / \text{velocity of approach}$$
$$= 2.40 / 4.00 = 0.60$$

After the collision



approach **4.00 m/s** · separation **2.40 m/s** · $e = 0.60$

Neither quantity belongs to one ball: the velocity of approach is how fast the **gap is closing**, the difference of the two velocities. Because e is a ratio of two speeds it has **no units**, and it cannot exceed 1 without energy coming from somewhere.

Coefficient of restitution

e = velocity of separation / velocity of approach

$$= (V_{Bf} - V_{Af}) / (V_{Ai} - V_{Bi})$$

Where V_{Bf} = final velocity of object B, V_{Af} = final velocity of object A, V_{Ai} = initial velocity of object A, V_{Bi} = initial velocity of object B.

- The coefficient of restitution is a ratio and therefore has no units.
- For perfectly elastic collision the COR is **1**
- For perfectly inelastic collisions the COR is **0**

Most rules for ball sports directly or indirectly specify what the coefficient of restitution must be for the ball on the specific playing surface or implement.

$$e = \frac{V_{Bf} - V_{Af}}{V_{Ai} - V_{Bi}}$$

velocity of separation over velocity of approach

$e = 1$ perfectly elastic — they separate as fast as they approached

$0 < e < 1$ real collisions — some kinetic energy is lost

$e = 0$ perfectly inelastic — they do not separate at all

The coefficient of restitution is affected by the nature of both objects. For ball sports the coefficient of restitution can be figured out using a simple equation.

If a ball is dropped from a specific height onto a fixed surface, then the height of the rebound and the drop height can be used to determine the coefficient of restitution. This is a measure of the bounciness of the ball and the implement or surface it strikes will greatly affect the outcome.

$$e = \sqrt{\frac{\text{bounce height}}{\text{drop height}}}$$

Elastic collisions

An elastic collision occurs when the two objects collide and then bounce off one another.

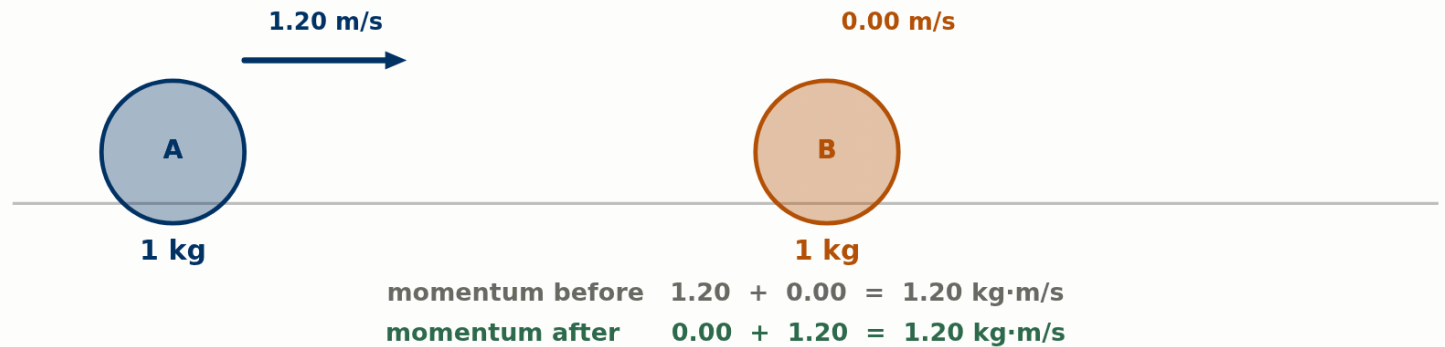
A collision between two pool or snooker balls is a good example of an almost totally elastic collision. The objects collide and then seem to bounce off of each other and move separately.

In a perfectly elastic collision, in a straight line, each object transfers all of its momentum to the other object.

$$m_A v_{A \text{ initial}} + m_B v_{B \text{ initial}} = m_A v_{A \text{ final}} + m_B v_{B \text{ final}}$$

before the collision

$e = 1.00$



A 1.20 → **0.00** m/s · B 0.00 → **1.20** m/s

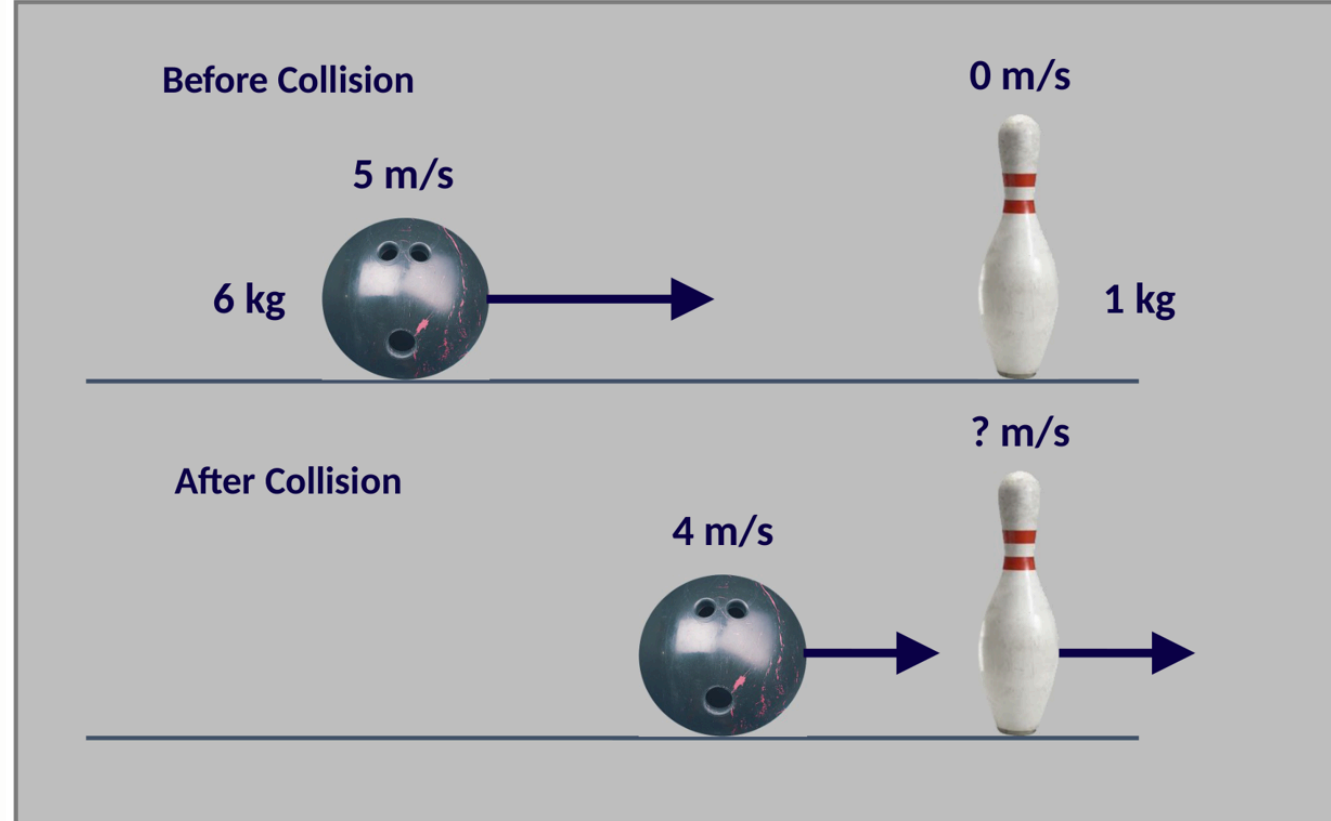
The total momentum is the same number before and after — that is the law, and it is one equation. It is not enough on its own: two unknowns need two equations, and the second is the coefficient of restitution.

▶ Collide

Example Problem

A 6 kg bowling ball enters an elastic collision with a 1 kg bowling pin. If the bowling ball has a velocity of 5 m/s before the collision and 4 m/s after the collision, what is the speed of the pin after the collision? (Assume that the speed of the pin before the collision is 0 m/s)





Solution

Before collision

After collision

$$(m_{\text{ball1}})(v_{\text{ball1}}) + (m_{\text{pin1}})(v_{\text{pin1}}) = (m_{\text{ball2}})(v_{\text{ball2}}) + (m_{\text{pin2}})(v_{\text{pin2}})$$

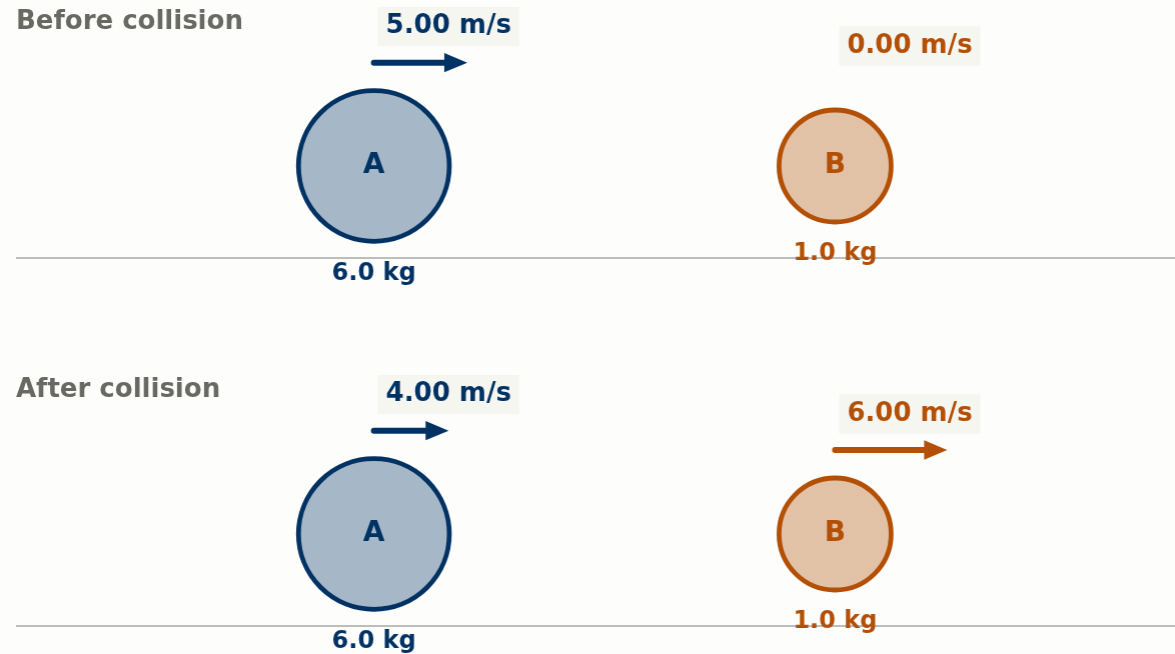
$$(6 \text{ kg})(5 \text{ m/s}) + (1 \text{ kg})(0 \text{ m/s}) = (6 \text{ kg})(4 \text{ m/s}) + (1 \text{ kg})(v_{\text{pin2}})$$

$$30 \text{ kg}\cdot\text{m/s} = 24 \text{ kg}\cdot\text{m/s} + v_{\text{pin2}}(1 \text{ kg})$$

$$v_{\text{pin2}} = \mathbf{6 \text{ m/s}}$$

After the collision, the pin has a velocity of 6 m/s in the original direction of the bowling ball.

The same collision, with the numbers on sliders



$e = 0.40$ · partly elastic

momentum is conserved

$$m_A v_{Ai} + m_B v_{Bi} = 30.00 \text{ kg}\cdot\text{m/s}$$

$$m_A v_{Af} + m_B v_{Bf} = 30.00 \text{ kg}\cdot\text{m/s}$$

the restitution equation

$$e = (v_{Bf} - v_{Af}) / (v_{Ai} - v_{Bi})$$
$$= 2.00 / 5.00 = 0.40$$

v_{Af}
4.00

v_{Bf}
6.00

KE kept
88%

A: 5.00 → 4.00 m/s · B: 0.00 → 6.00 m/s

Momentum is the same before and after, always — that is the law. What the coefficient of restitution decides is how it is **shared out**. Between the two, some kinetic energy is lost but they still separate.

bowling ball

equal, B at rest

equal, head-on

2 kg meets 1 kg

inelastic

Andre & Hulk

Elastic collisions

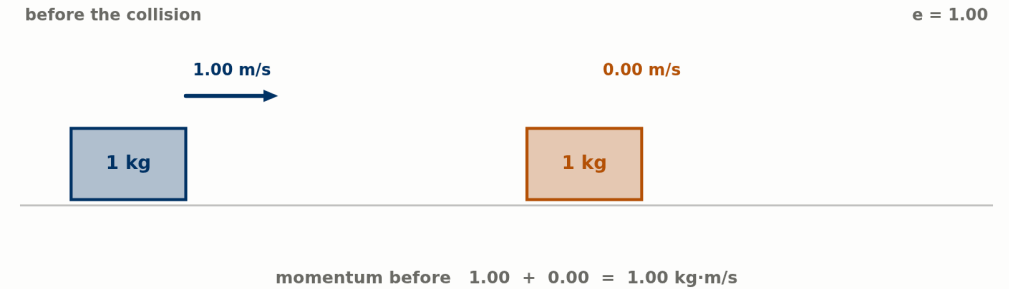
Equal masses, B at rest

Since m is the same we can take it out of the equation. Also assume $v = 1$ m/s and $v_{B \text{ initial}}$ is zero.

$$m_A v_{A \text{ initial}} + m_B v_{B \text{ initial}} = m_A v_{A \text{ final}} + m_B v_{B \text{ final}}$$

$$(1 \text{ m/s}) + 0 \text{ m/s} = v_{A \text{ final}} + v_{B \text{ final}}$$

$$v_{A \text{ final}} = 1 - v_{B \text{ final}}$$



A $1.00 \rightarrow ?$ m/s · B $0.00 \rightarrow ?$ m/s

Momentum gives you one equation and there are two unknowns in it, so it cannot be solved yet. The second equation is the coefficient of restitution, on the next slide.

▶ Collide

Elastic collisions

Equal masses, B at rest — solution

Also we know that the coefficient of restitution is 1.

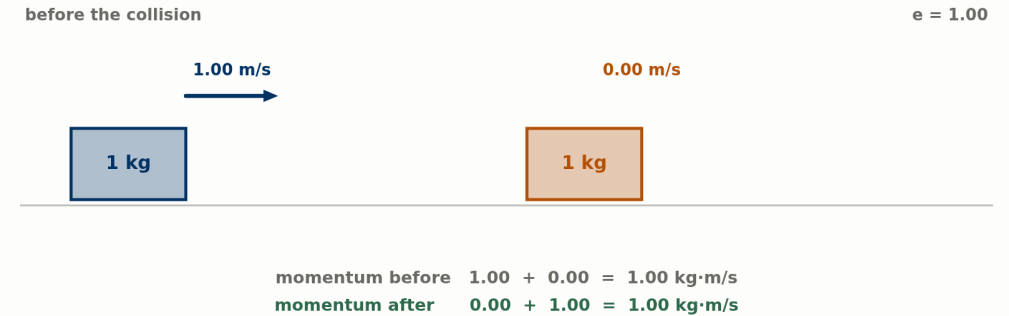
$$e = (v_{Bf} - v_{Af}) / (v_{Ai} - v_{Bi})$$

$$1 = (v_{Bf} - (1 - v_{B \text{ final}})) / ((1 \text{ m/s}) - 0 \text{ m/s})$$

$$1 = -1 + 2v_{Bf}$$

$$v_{Bf} = \mathbf{1 \text{ m/s}}$$

Sub v_{Bf} back into equation 1: $v_{A \text{ final}} = 1 - 1 = \mathbf{0 \text{ m/s}}$



A 1.00 → **0.00** m/s · B 0.00 → **1.00** m/s

The total momentum is the same number before and after — that is the law, and it is one equation. It is not enough on its own: two unknowns need two equations, and the second is the coefficient of restitution.

▶ Collide

Elastic collisions

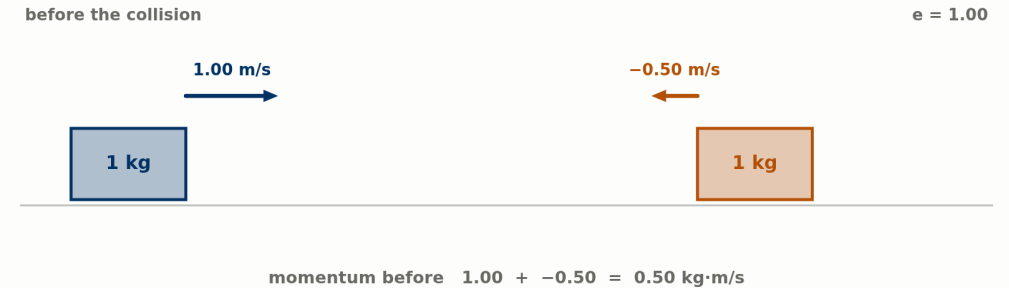
Equal masses, head on

Since m is the same we can take it out of the equation. Also assume $v = 1$ m/s, with B coming the other way at 0.5 m/s.

$$m_A v_{A \text{ initial}} + m_B v_{B \text{ initial}} = m_A v_{A \text{ final}} + m_B v_{B \text{ final}}$$

$$1 \text{ m/s} - 0.5 \text{ m/s} = v_{A \text{ final}} + v_{B \text{ final}}$$

$$v_{A \text{ final}} = 0.5 - v_{B \text{ final}}$$



A 1.00 → ? m/s · B -0.50 → ? m/s

Momentum gives you one equation and there are two unknowns in it, so it cannot be solved yet. The second equation is the coefficient of restitution, on the next slide.

▶ Collide

Elastic collisions

Equal masses, head on — solution

Also we know that the coefficient of restitution is 1.

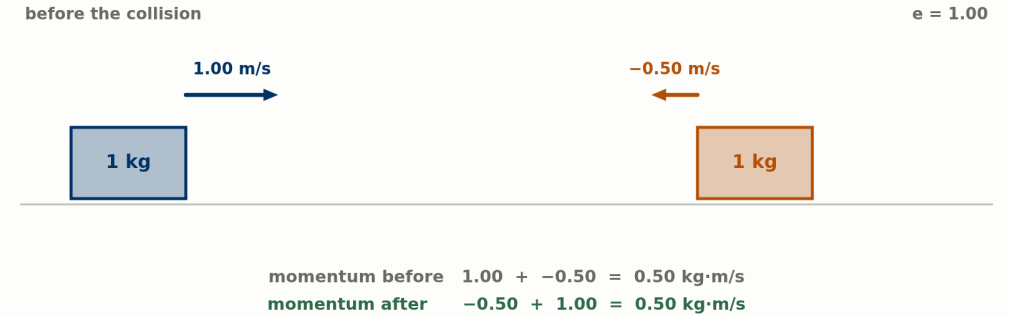
$$e = (v_{Bf} - v_{Af}) / (v_{Ai} - v_{Bi})$$

$$1 = (v_{Bf} - (0.5 - v_{B \text{ final}})) / ((1 \text{ m/s}) - (-0.5 \text{ m/s}))$$

$$1 = (-0.5 + 2v_{Bf}) / 1.5$$

$$1.5 = -0.5 + 2v_{Bf} \rightarrow v_{Bf} = \mathbf{1 \text{ m/s}}$$

$$v_{A \text{ final}} = 0.5 - 1 = \mathbf{-0.5 \text{ m/s}}$$



A 1.00 → **-0.50** m/s · B -0.50 → **1.00** m/s

The total momentum is the same number before and after — that is the law, and it is one equation. It is not enough on its own: two unknowns need two equations, and the second is the coefficient of restitution.

▶ Collide

Elastic collisions

Different masses

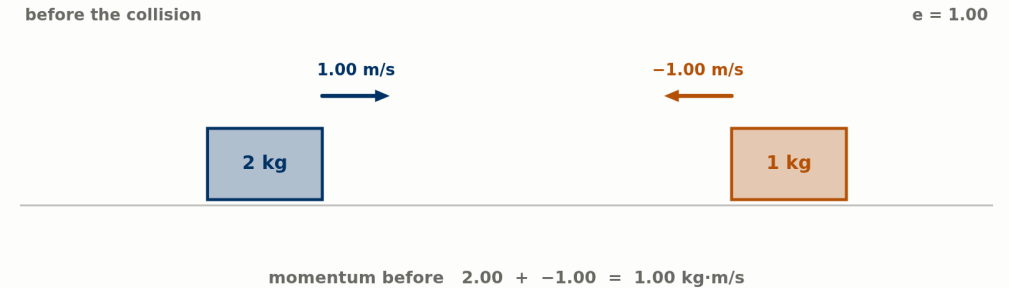
Assume that $m = 1$ kg and $v = 1$ m/s. A is twice the mass of B and they are coming straight at each other.

$$m_A v_{A \text{ initial}} + m_B v_{B \text{ initial}} = m_A v_{A \text{ final}} + m_B v_{B \text{ final}}$$

$$(2 \text{ kg})(1 \text{ m/s}) + (1 \text{ kg})(-1 \text{ m/s}) = (2 \text{ kg})v_{A \text{ final}} + (1 \text{ kg})v_{B \text{ final}}$$

$$1 = 2v_{A \text{ final}} + v_{B \text{ final}}$$

$$v_{B \text{ final}} = 1 - 2v_{A \text{ final}}$$



A $1.00 \rightarrow ?$ m/s · B $-1.00 \rightarrow ?$ m/s

Momentum gives you one equation and there are two unknowns in it, so it cannot be solved yet. The second equation is the coefficient of restitution, on the next slide.

▶ Collide

Elastic collisions

Different masses — solution

Also we know that the coefficient of restitution is 1.

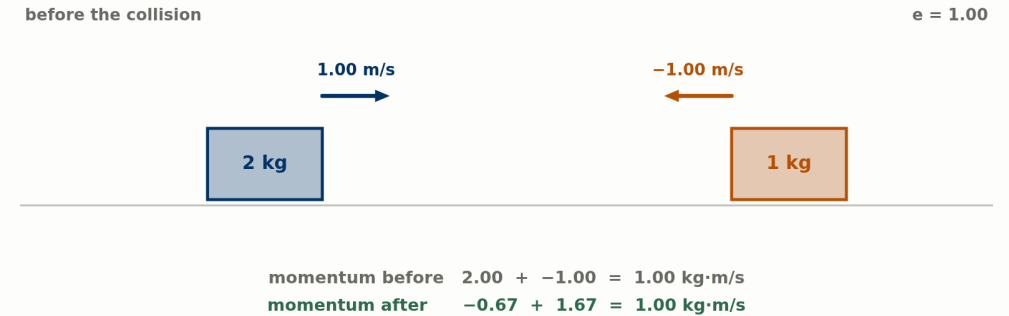
$$e = (v_{Bf} - v_{Af}) / (v_{Ai} - v_{Bi})$$

$$1 = ((1 - 2v_{A \text{ final}}) - v_{Af}) / ((1 \text{ m/s}) - (-1 \text{ m/s}))$$

$$1 = (1 - 3v_{Af}) / 2$$

$$v_{Af} = -0.33 \text{ m/s}$$

$$v_{B \text{ final}} = 1 - 2(-0.33 \text{ m/s}) = 1.66 \text{ m/s}$$



A 1.00 → **-0.33** m/s · B -1.00 → **1.67** m/s

The total momentum is the same number before and after — that is the law, and it is one equation. It is not enough on its own: two unknowns need two equations, and the second is the coefficient of restitution.

▶ Collide

Inelastic collisions

An inelastic collision occurs when two objects collide and they behave as one object after the collision.

A common example of a perfectly inelastic collision is when two objects collide and then stick together afterwards. This equation describes the conservation of momentum for an inelastic collision:

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_{\text{combined}}$$

Momentum is still conserved. Kinetic energy is not — it has gone into deformation, sound and heat.

before the collision

$e = 0.00$



momentum before	1.00	+	0.00	=	1.00 kg·m/s
momentum after	0.50	+	0.50	=	1.00 kg·m/s

A 1.00 → **0.50** m/s · B 0.00 → **0.50** m/s

The total momentum is the same number before and after — that is the law, and it is one equation. It is not enough on its own: two unknowns need two equations, and the second is the coefficient of restitution.

▶ Collide

Inelastic collisions

Equal masses, B at rest

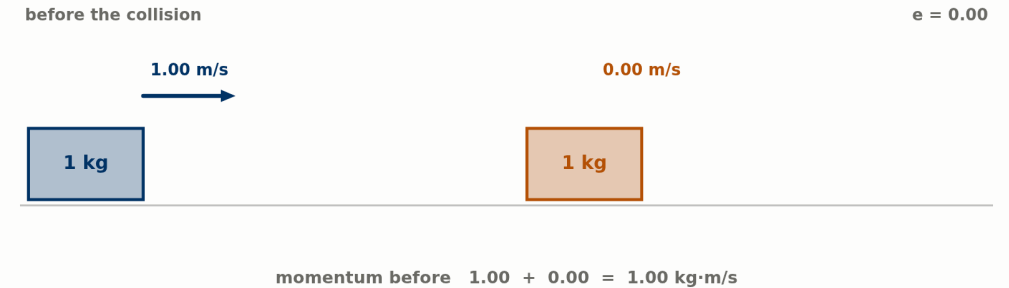
Assume that $m = 1 \text{ kg}$ and $v = 1 \text{ m/s}$.

$$m_A v_{A \text{ initial}} + m_B v_{B \text{ initial}} = m_A v_{A \text{ final}} + m_B v_{B \text{ final}}$$

$$(1 \text{ kg})(1 \text{ m/s}) + (1 \text{ kg})(0 \text{ m/s}) = (1 \text{ kg})v_{1 \text{ final}} + (1 \text{ kg})v_{2 \text{ final}}$$

$$1 = v_{A \text{ final}} + v_{B \text{ final}}$$

$$v_{B \text{ final}} = 1 - v_{A \text{ final}}$$



A 1.00 → ? m/s · B 0.00 → ? m/s

Momentum gives you one equation and there are two unknowns in it, so it cannot be solved yet. The second equation is the coefficient of restitution, on the next slide.

▶ Collide

Inelastic collisions

Equal masses, B at rest — solution

Also we know that the coefficient of restitution is 0.

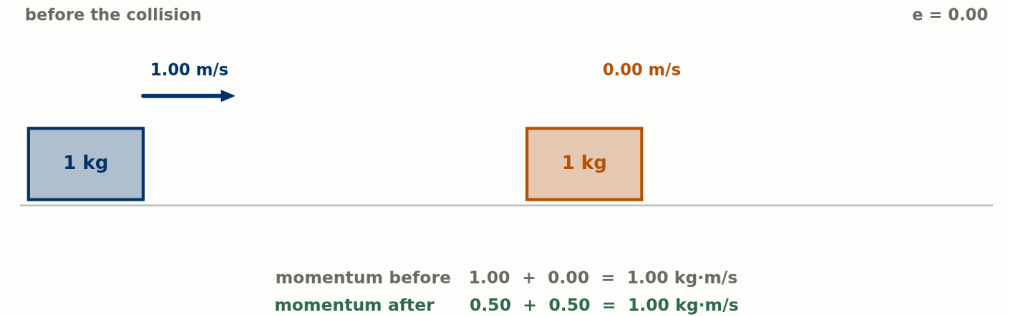
$$e = (v_{Bf} - v_{Af}) / (v_{Ai} - v_{Bi})$$

$$0 = ((1 - v_{A \text{ final}}) - v_{Af}) / ((1 \text{ m/s}) - (0 \text{ m/s}))$$

$$0 = 1 - 2v_{Af}$$

$$v_{Af} = \mathbf{0.5 \text{ m/s}}$$

$$v_{B \text{ final}} = 1 - 0.5 \text{ m/s} = \mathbf{0.5 \text{ m/s}}$$



A 1.00 → **0.50** m/s · B 0.00 → **0.50** m/s

The total momentum is the same number before and after — that is the law, and it is one equation. It is not enough on its own: two unknowns need two equations, and the second is the coefficient of restitution.

▶ Collide

All four cases, with the numbers on sliders



$e = 1.00$ · perfectly elastic

momentum is conserved

$$m_A v_{Ai} + m_B v_{Bi} = \mathbf{1.00} \text{ kg}\cdot\text{m/s}$$

$$m_A v_{Af} + m_B v_{Bf} = \mathbf{1.00} \text{ kg}\cdot\text{m/s}$$

the restitution equation

$$e = (v_{Bf} - v_{Af}) / (v_{Ai} - v_{Bi})$$
$$= 1.00 / 1.00 = \mathbf{1.00}$$

v_{Af}
0.00

v_{Bf}
1.00

KE kept
100%

A: 1.00 → **0.00** m/s · B: 0.00 → **1.00** m/s

Momentum is the same before and after, always — that is the law. What the coefficient of restitution decides is how it is **shared out**. At $e = 1$ no kinetic energy is lost at all.

bowling ball

equal, B at rest

equal, head-on

2 kg meets 1 kg

inelastic

Andre & Hulk

Example Problem

During the feature match of Wrestlemania III, Hulk Hogan and Andre the Giant run towards each other and collide. If Andre has a mass of 235 kg and was traveling 1.4 m/s, and if Hulk has a mass of 115 kg and was traveling at -3.8 m/s, what is their combined velocity? (assuming that they entangle and continue traveling as one)



Solution

- given in this question:

$$m_1 = 235 \text{ kg} \quad m_2 = 115 \text{ kg}$$
$$v_1 = 1.4 \text{ m/s} \quad v_2 = -3.8 \text{ m/s}$$

- Momentum Before Collision = Momentum After Collision

$$(mv)_1 + (mv)_2 = (m_1 + m_2)v$$

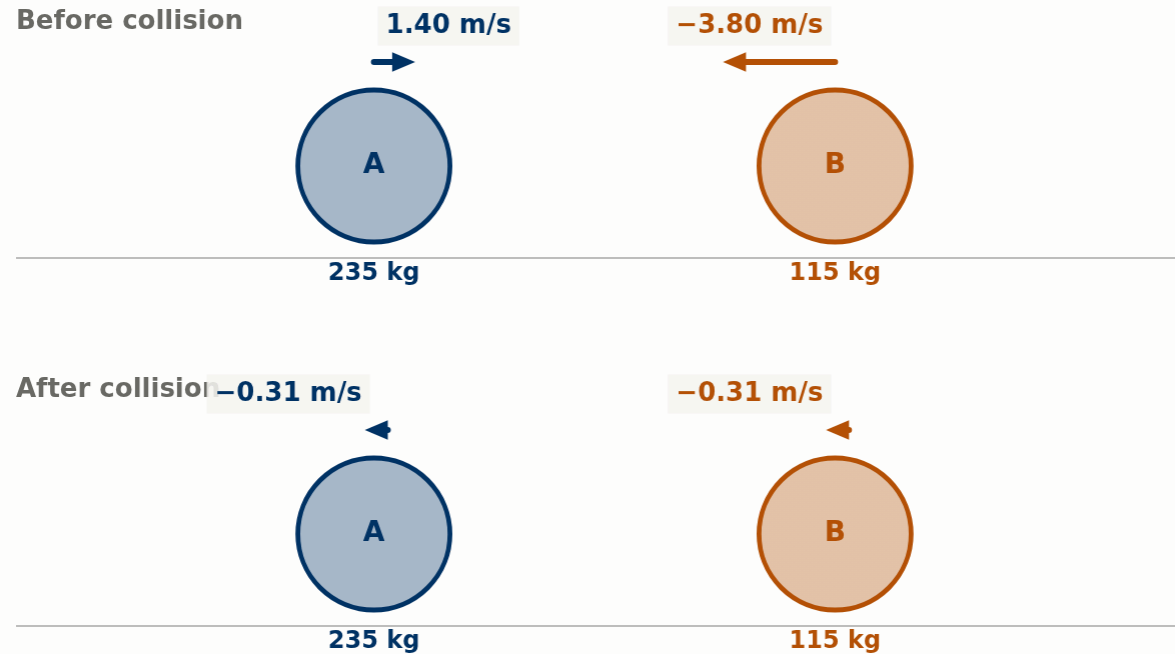
$$(235 \text{ kg})(1.4 \text{ m/s}) + (115 \text{ kg})(-3.8 \text{ m/s}) = (235 \text{ kg} + 115 \text{ kg})v$$

$$329 \text{ kg}\cdot\text{m/s} - 437 \text{ kg}\cdot\text{m/s} = (350 \text{ kg})(v)$$

$$v = \mathbf{-0.3 \text{ m/s}}$$

Their combined velocity is 0.3 m/s in Hulk Hogan's original direction

The same collision, with the numbers on sliders



$e = 0.00$ · perfectly inelastic

momentum is conserved

$$m_A v_{Ai} + m_B v_{Bi} = -108.00 \text{ kg}\cdot\text{m/s}$$

$$m_A v_{Af} + m_B v_{Bf} = -108.00 \text{ kg}\cdot\text{m/s}$$

the restitution equation

$$e = (v_{Bf} - v_{Af}) / (v_{Ai} - v_{Bi})$$
$$= 0.00 / 5.20 = 0.00$$

v_{Af}
-0.31

v_{Bf}
-0.31

KE kept
2%

A: 1.40 → -0.31 m/s · B: -3.80 → -0.31 m/s

Momentum is the same before and after, always — that is the law. What the coefficient of restitution decides is how it is **shared out**. At $e = 0$ they leave together at one velocity and the lost kinetic energy has gone into deformation and heat.

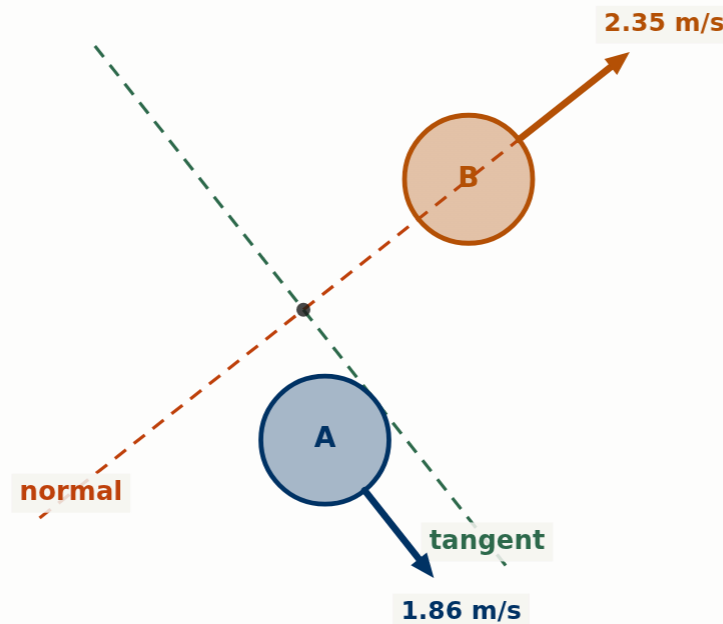
Linear Kinetics

You are required to be able to solve questions for conservation of momentum only on a **straight line** — 1 dimension.

However you will be expected to **conceptually understand** simple collisions in 2 dimensions.

Two smooth objects can only push each other along the **line joining their centres**. Everything in the rest of this lecture follows from that one sentence.

each ball leaves along its own new line



Collisions in two dimensions

The problem

For elastic collisions in two dimensions we need to break the momentum into components. Each ball has a mass of **0.1 kg** and ball B was at rest. What is the speed and direction of ball B after the collision?

$$P_o = P_f$$

$$P_{ox} = P_{fx} , P_{oy} = P_{fy}$$



B leaves at **3.45 m/s**, 16.9° below the x axis

The y momentum started at zero, so whatever A takes upward, B must take downward by exactly as much. Two equations, one for each axis — that is the whole of two-dimensional conservation.

1 · the set-up

2 · A leaves

3 · components

4 · the answer

Collisions in two dimensions

Momentum in x

A leaves at 2 m/s, 30° above its original line. Write the x equation and the only unknown left in it is B's x component.

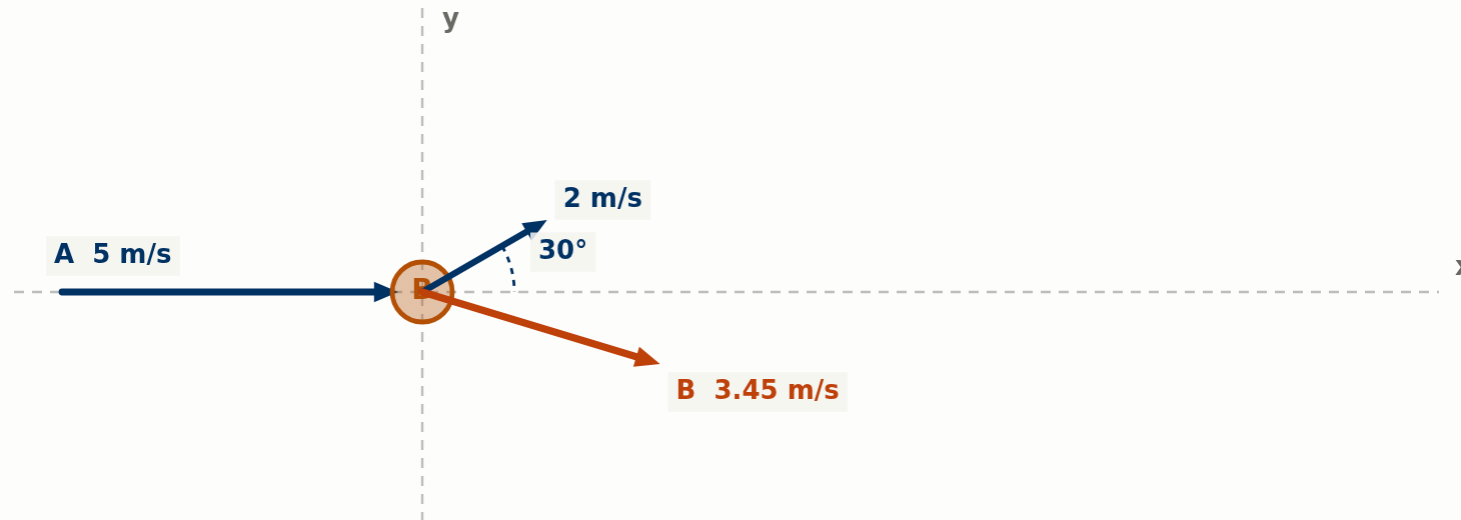
$$P_{ox} = P_{fx}$$

$$m_A v_{Aox} + m_B v_{Box} = m_A v_{Afx} + m_B v_{Bfx}$$

$$(0.1 \text{ kg})(5 \text{ m/s}) + (0.1 \text{ kg})(0 \text{ m/s}) = (0.1 \text{ kg})(2 \text{ m/s} \cdot \cos 30) + (0.1 \text{ kg})(v_{Bfx})$$

$$0.5 + 0 = 0.17 + 0.1 v_{Bfx}$$

$$v_{Bfx} = \mathbf{3.3 \text{ m/s}}$$



Momentum in y

Now the same thing along y. Nothing was moving in y before the collision, so the whole of that side of the equation is zero.

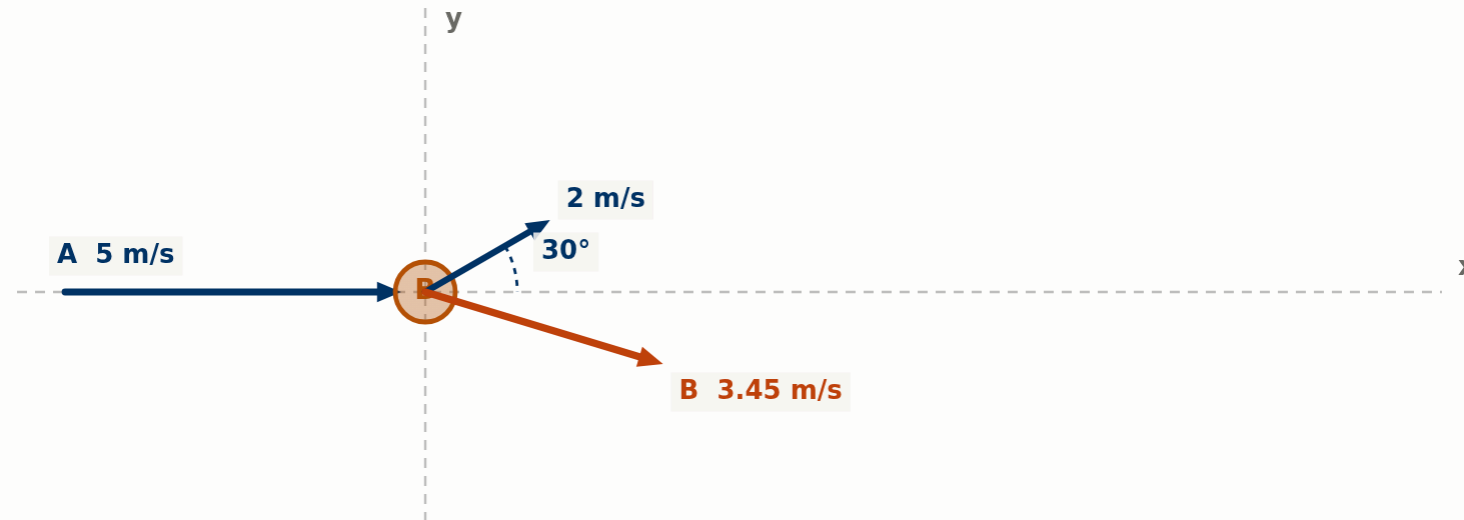
$$P_{oy} = P_{fy}$$

$$m_A v_{Aoy} + m_B v_{Boy} = m_A v_{Afy} + m_B v_{Bfy}$$

$$(0.1 \text{ kg})(0 \text{ m/s}) + (0.1 \text{ kg})(0 \text{ m/s}) = (0.1 \text{ kg})(2 \text{ m/s} \sin 30^\circ) + (0.1 \text{ kg})(v_{Bfy})$$

$$0 = 0.1 + 0.1 v_{Bfy}$$

$$v_{Bfy} = -1 \text{ m/s}$$



Magnitude and direction

Two components make one vector. Pythagoras for the size of it, and the arctangent for which way it points.

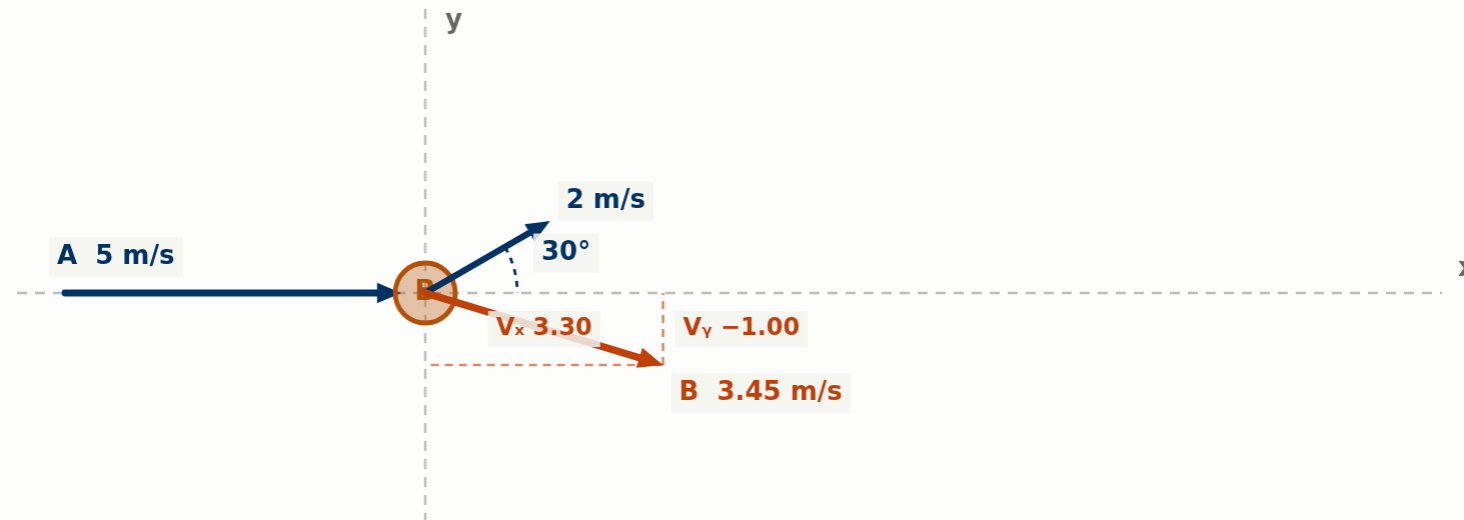
$$V_{bf}^2 = V_x^2 + V_y^2$$

$$V_{bf}^2 = (3.3 \text{ m/s})^2 + (-1 \text{ m/s})^2$$

$$V_{bf} = \mathbf{3.45 \text{ m/s}}$$

$$\tan\theta = V_y / V_x = -1 / 3.3$$

$$\theta = \tan^{-1}(-0.303) = \mathbf{16.85^\circ \text{ below the x axis}}$$

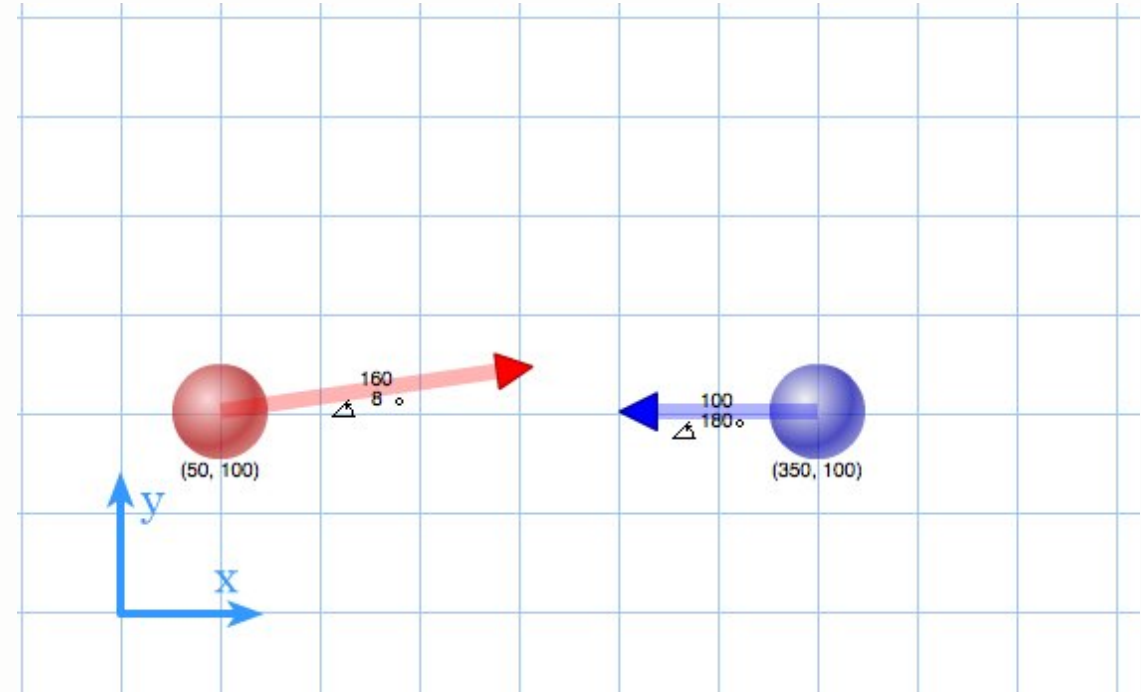


B leaves at **3.45 m/s**, 16.9° below the x axis

Linear Kinetics

In the previous example we had an idea of the angle of the balls velocities before and after the collision. This allowed us to determine the magnitude of the velocity. However that is not always the case.

To determine the angle and magnitude of the balls velocities before and after the collision we need to set up a frame of reference and explore the conservation of momentum in axes **normal** (perpendicular) and **tangent** (parallel) to the contact surface interaction.

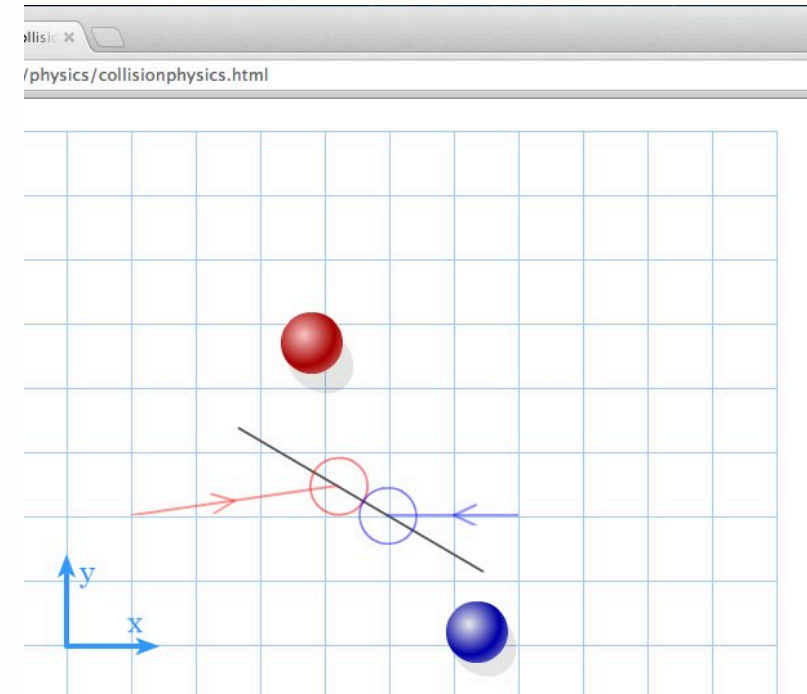


Normal and tangent axes

With our original x and y axis you can see:

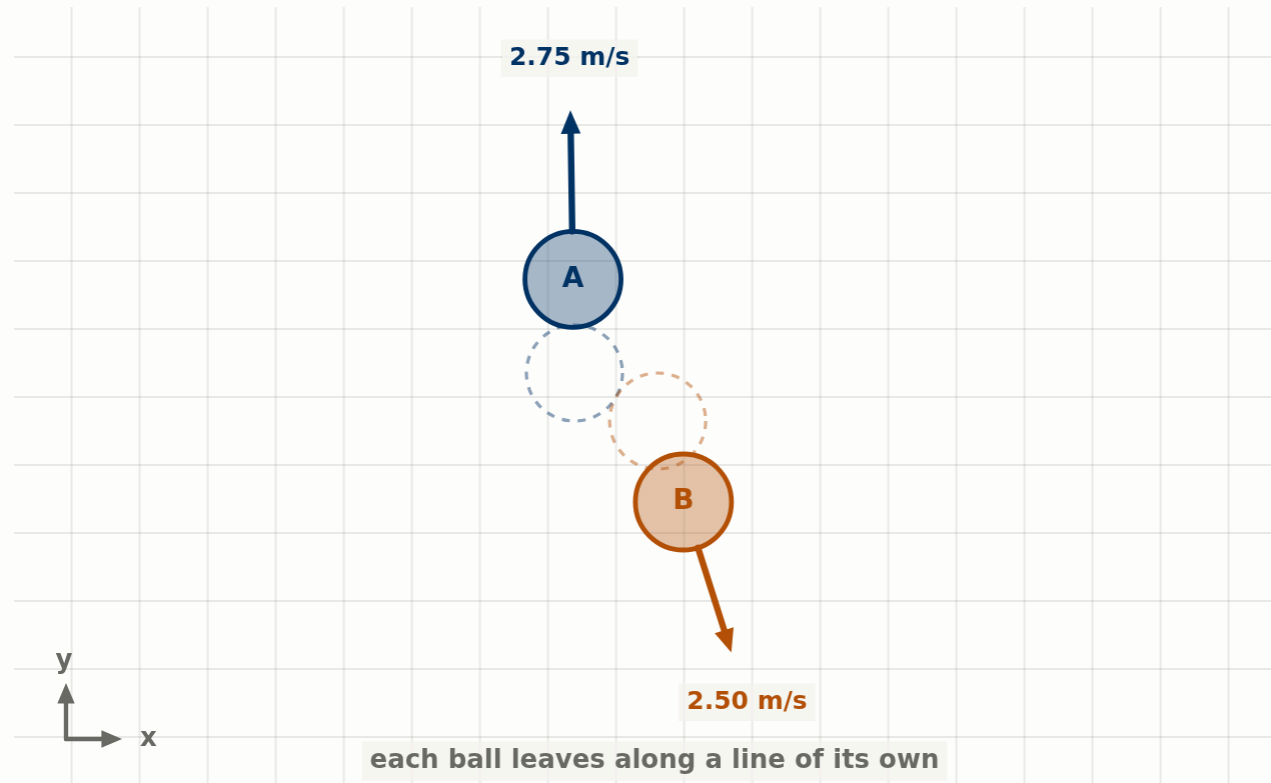
- the velocities of the balls before the collision
- The axis **normal** (perpendicular) to the contacting surfaces
- The axis **tangent** (parallel) to the contacting surfaces

Neither of those two axes lines up with x or y, which is exactly the problem.



Normal and tangent axes

The collision, written down twice



line of centres -30° to x · they leave at 2.75 and 2.50 m/s

Two smooth balls can only push on each other one way: along the line joining their centres. On **x** and **y** both balls have two components, all four of them change, and there is nothing to hold on to. On the **normal** and the **tangent** it comes apart: along the tangent nothing happens at all, and along the normal it is the one-dimensional collision from the start of the lecture.

▶ Collide

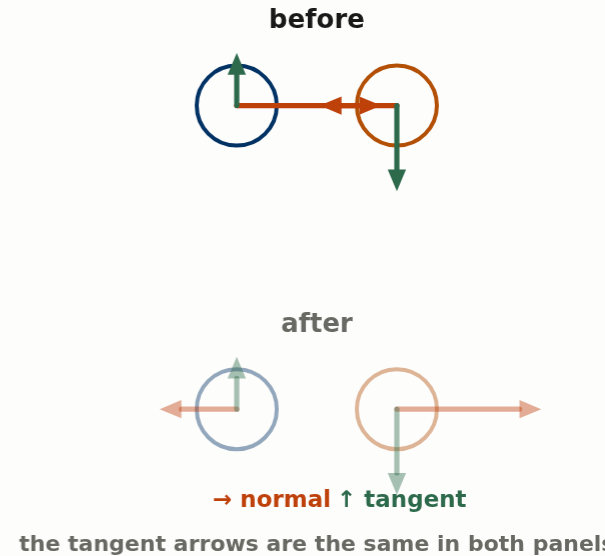
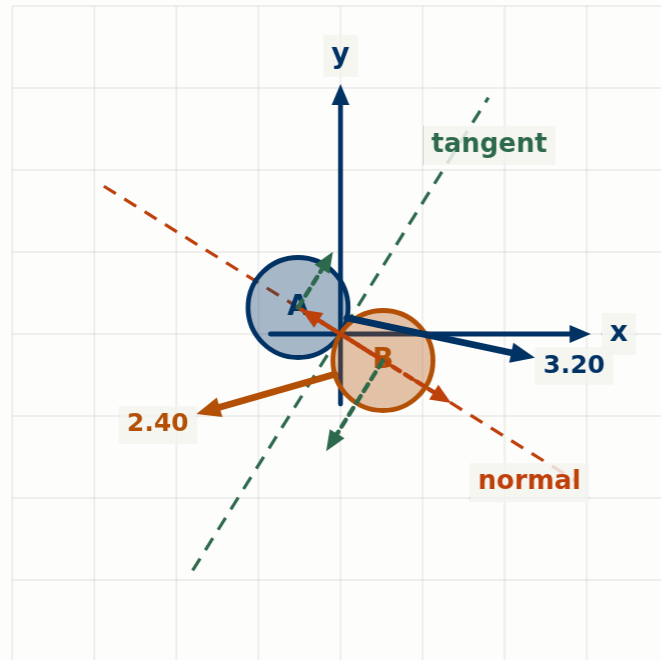
the collision

on x, y

on n, t

Turning the frame of reference

Turn the whole frame — grid, axes, velocities and all — until the line joining the centres is horizontal.



frame turned 0° of 32°

components before

normal A 3.01 B -1.61

tangent A 1.09 B -1.78

components after

normal A -1.61 B 3.01

tangent A 1.09 B -1.78

A after
1.94

B after
3.50

the original **x, y** frame · before the collision

Nothing about the collision changes while that slider moves — the balls and their velocities are doing exactly what they were doing. What changes is the pair of axes you describe them with. Choose **x** and **y** and both balls have two components that both change, and you are stuck. Choose the **normal** and the **tangent** and the problem separates: the tangential components come through the collision **untouched** — first law, one component at a time — and everything that happens happens on the normal, where it is the same one-dimensional problem as the start of the lecture.

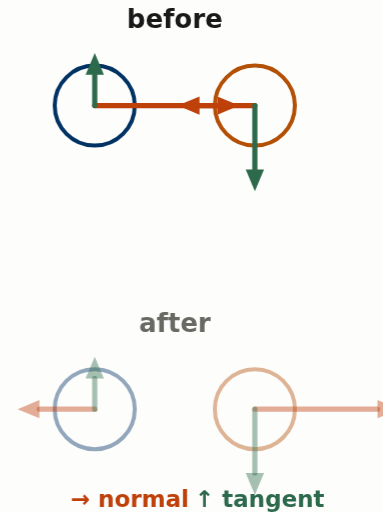
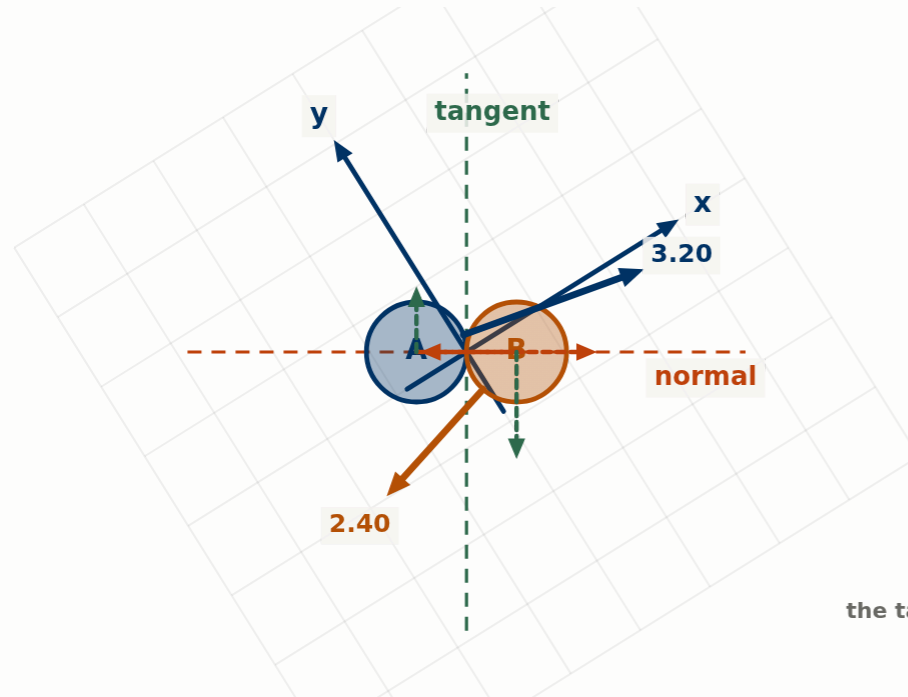
▶ Turn the frame

before

after

Components in the new frame

Then determine the **normal** and **tangent** component vectors *before* the collision.



the tangent arrows are the same in both panel!

frame turned 32° of 32°

components before

normal A 3.01 B -1.61

tangent A 1.09 B -1.78

components after

normal A -1.61 B 3.01

tangent A 1.09 B -1.78

A after
1.94

B after
3.50

the **normal**–**tangent** frame · before the collision

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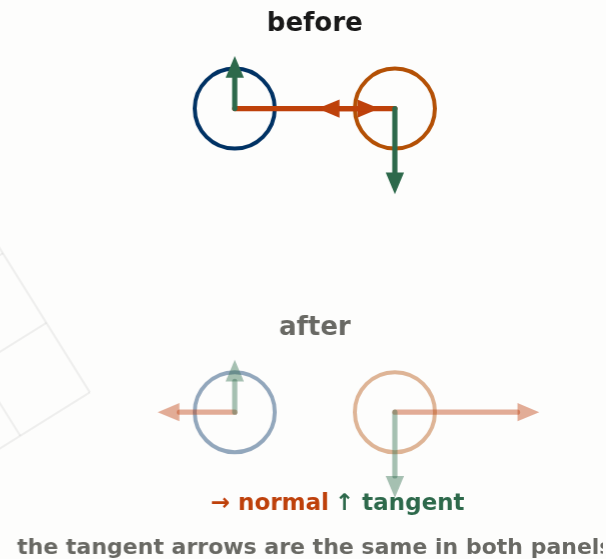
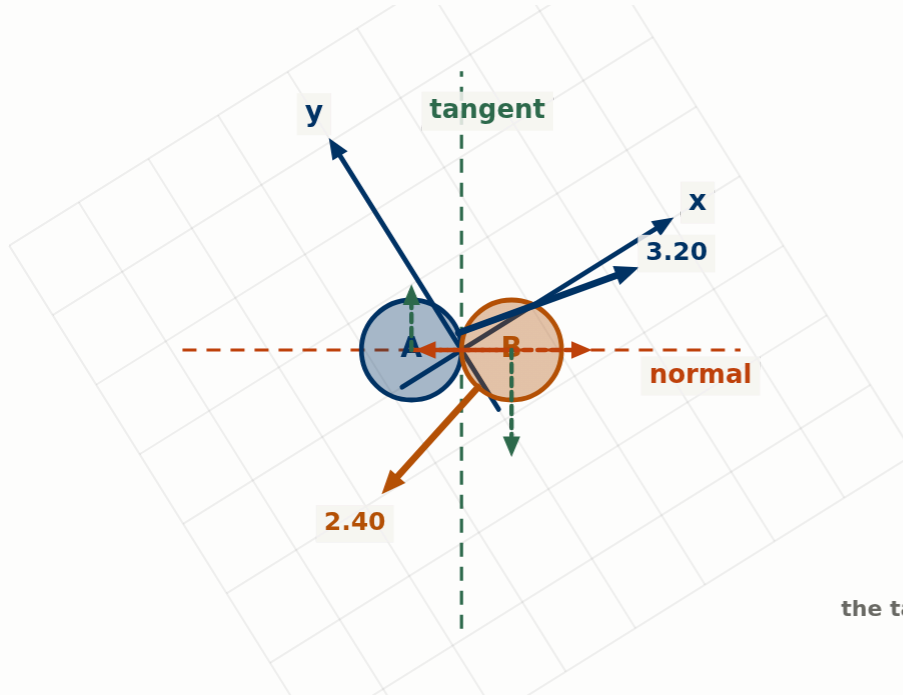
▶ Collide

before

after

After the collision

The **tangential** components remain **unchanged**; conservation of linear momentum is maintained for the **normal** components.



frame turned 32° of 32°

components before

normal A 3.01 B -1.61

tangent A 1.09 B -1.78

components after

normal A -1.61 B 3.01

tangent A 1.09 B -1.78

A after
1.94

B after
3.50

the **normal**–**tangent** frame · before the collision

Nothing about the collision changes while that slider moves — the balls and their velocities are doing exactly what they were doing. What changes is the pair of axes you describe them with. Choose x and y and both balls have two components that both change, and you are stuck. Choose the **normal** and the **tangent** and the problem separates: the tangential components come through the collision **untouched** — first law, one component at a time — and everything that happens happens on the normal, where it is the same one-dimensional problem as the start of the lecture.

▶ Collide

before

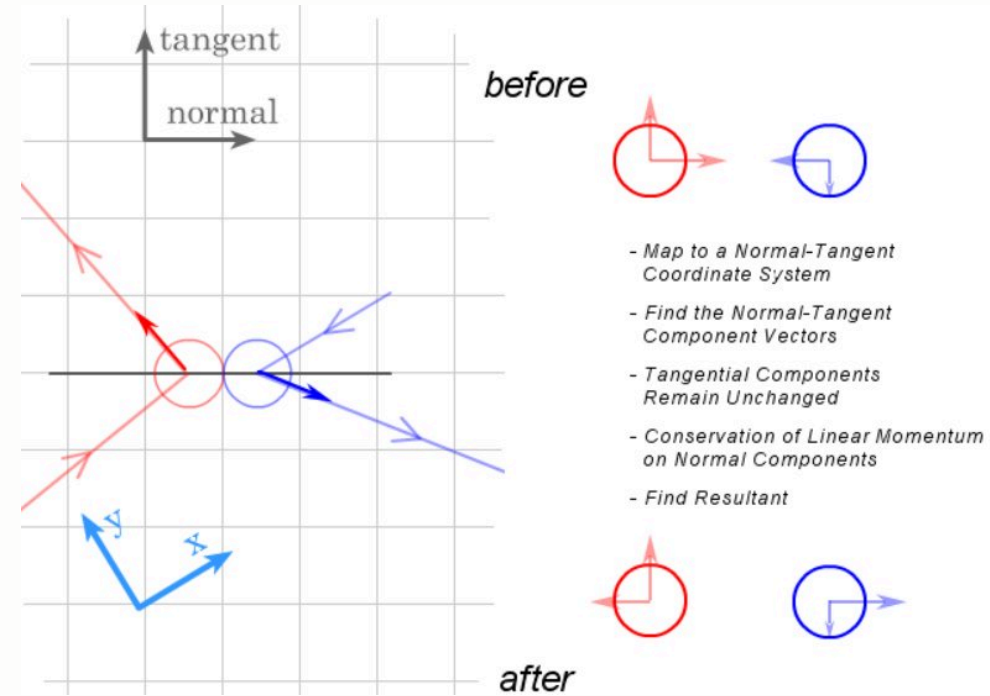
after

Linear Kinetics

Let's think about that in terms of Newton's 1st law.

An object at rest tends to stay at rest and an object in motion tends to stay in motion (**in its component direction**) unless a net external force acts upon it.

- 1 Map to a normal–tangent coordinate system
- 2 Find the normal–tangent component vectors
- 3 Tangential components remain unchanged
- 4 Conservation of linear momentum on normal components
- 5 Find the resultant



Transfer of momentum

Understanding Momentum is really important when dealing with the combined motion of the human body and techniques to **transfer momentum from one limb to another.**

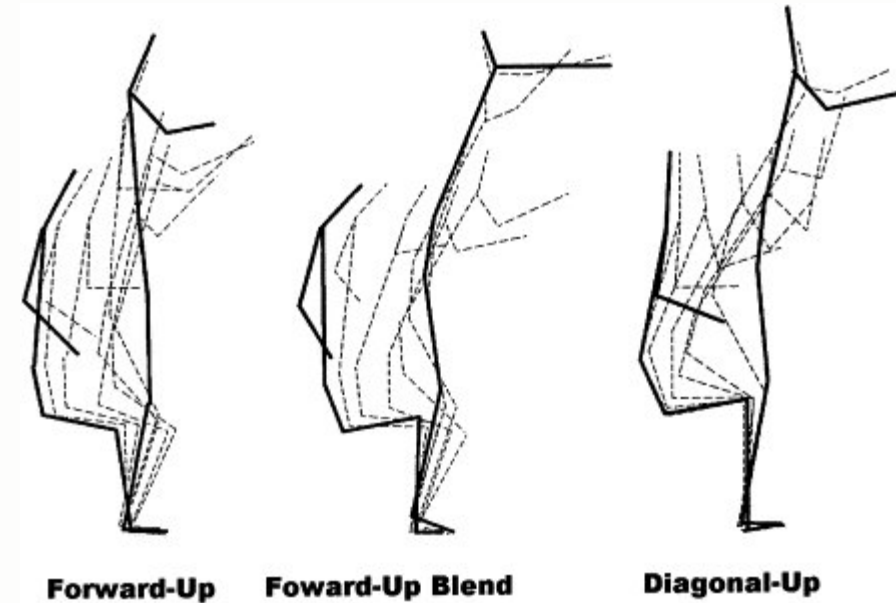
The outcome of transferring momentum from one limb to another can be explained by the conservation of momentum of the system.



Transfer of momentum in sit to stand

The transfer of momentum during human movement can be discussed in reference to simple and complex tasks.

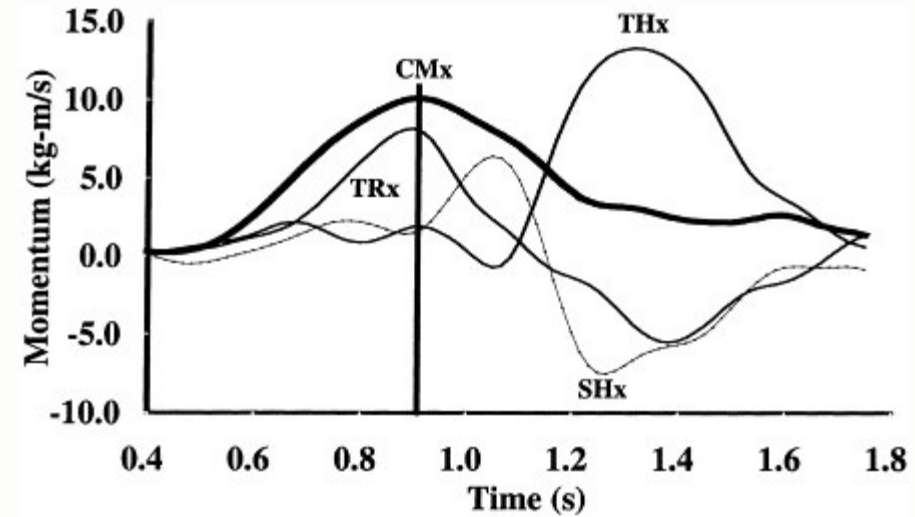
In a sit to stand task we use transfer of momentum to help us stand. By leaning forward our trunk has momentum which can be used to assist in hip and leg extension.



Horizontal momentum during sit to stand

Dark vertical line indicates time of occurrence of peak horizontal momentum of the CM.

- CMx is the horizontal momentum of whole body's CM
- TRx is the horizontal momentum of the trunk
- THx is the horizontal momentum of the thigh
- SHx is the horizontal momentum of the shank

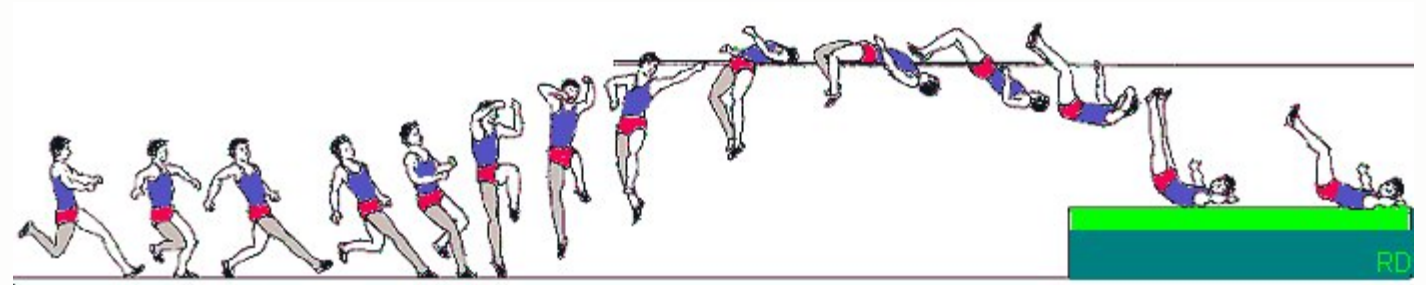


Transfer of momentum in the high jump

During a complex task such as high jump we use transfer of momentum to help us increase our **vertical velocity**.

The free limbs contribute by transferring their individual momentum to the velocity of the whole body:

- Contralateral arm relative momentum
- Ipsilateral arm relative momentum
- Lead leg relative momentum
- Combined free limb relative momentum



$$CARM = m_{CA} V_{CA/CS}$$

$$IARM = m_{IA} V_{IA/IS}$$

$$LLRM = m_{LL} V_{LL/H}$$

Free limb relative momentum

- This shows that the RM of the arms are negative at touch-down as the arms are being driven downwards
- the leg is flexing at the knee giving its CM a net upward velocity
- The RM of the leg continues to increase as it is flexed and elevated
- The RM of arms becomes positive as they reach the bottom of their swing and are elevated
- The RM of each arm are similar and they reach their peak at approximately the same point in time but that of the lead leg reaches its peak earlier than the arms

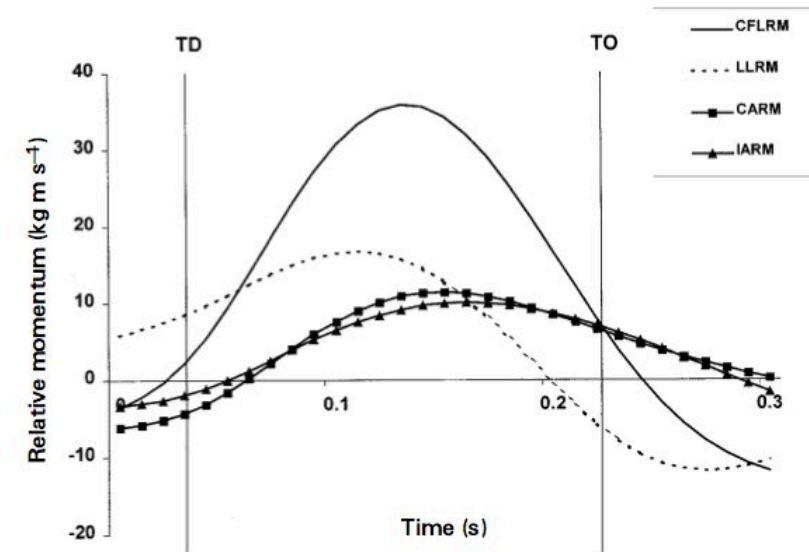


Figure 1. Vertical relative momentum for individual free limbs (CARM= contralateral arm relative momentum, IARM= ipsilateral arm relative momentum, LLRM= lead leg relative momentum) and combined free limbs relative momentum (CFLRM) for athlete Austin.

Free limb relative momentum

The combined free limb RM (CFLRM) reaches a peak just after the lead leg but just before the arms. The relative momentum of all free limbs **reduces at take-off**, illustrating the transfer of each segment momentum to the rest of the body.

The graph shows the estimated additional vertical force generated as a result of the transfer of momentum.

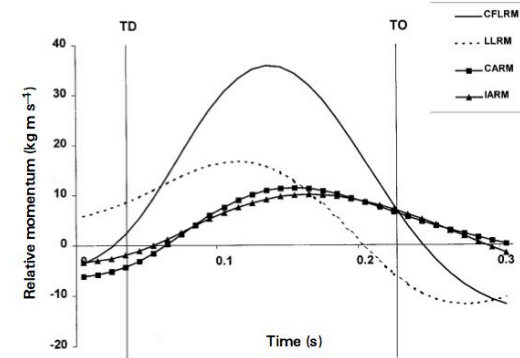


Figure 1. Vertical relative momentum for individual free limbs (CARM= contralateral arm relative momentum, IARM= ipsilateral arm relative momentum, LLRM= lead leg relative momentum) and combined free limbs relative momentum (CFLRM) for athlete Austin.

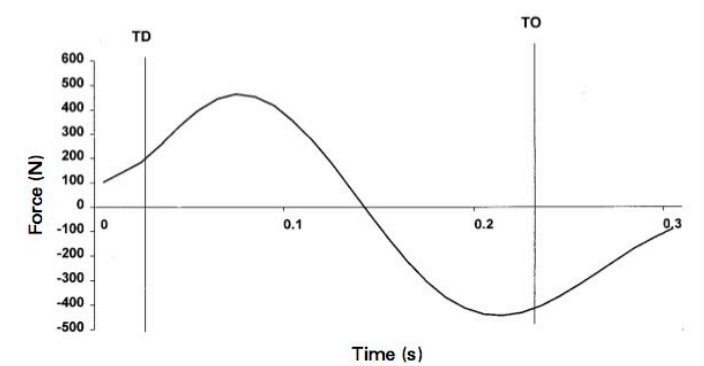


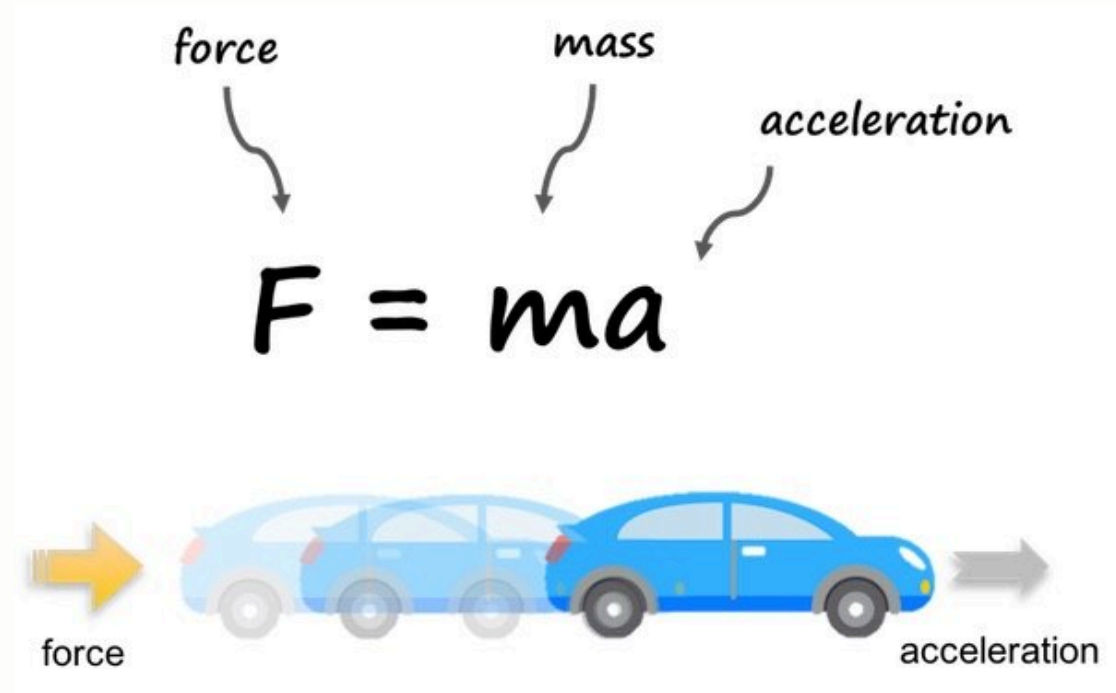
Figure 4. Typical graph of the estimated additional force generated as a result of limb motion.

2nd Law: The Law of Acceleration

Law II

“A force applied to a body causes an acceleration of that body of a magnitude proportional to the force, in the direction of the force, and inversely proportional to the body’s mass.”

$$F = ma$$



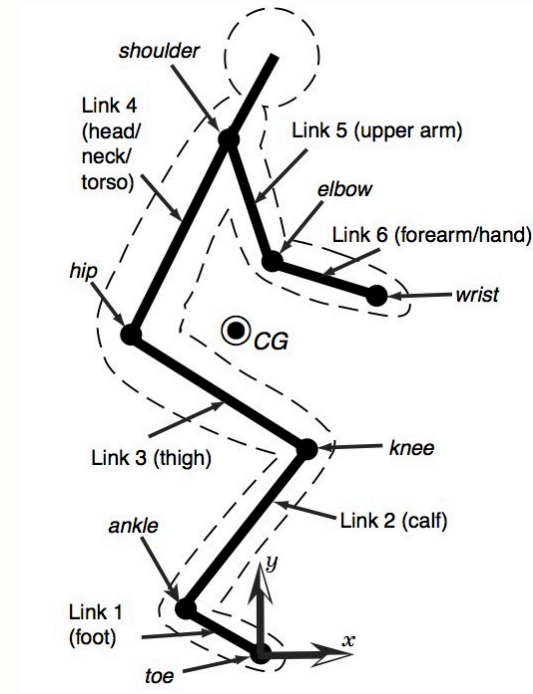
2nd Law: The Law of Acceleration

Newton's second law gives us the connection between **force (F)** and **motion (a)**.

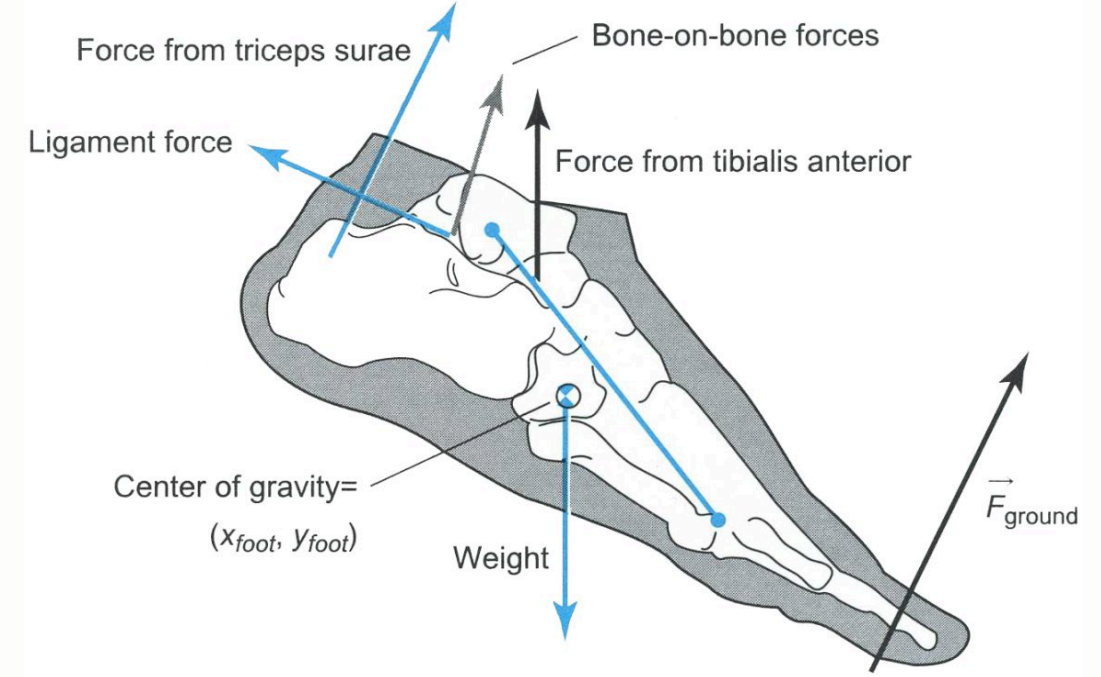
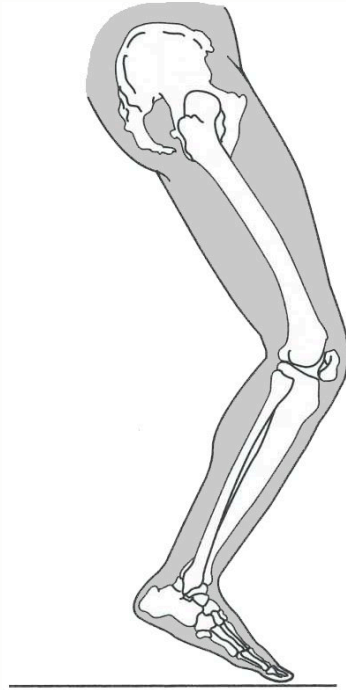
It provides a framework for thinking about how all of the forces (internal and external) can control motion.

We will use this law as a basis for our discussion of static and dynamic analysis including **inverse dynamics**.

$$F = ma$$



Free Body Diagram



3rd Law: The Law of Reaction

Law III

“Every applied force is accompanied by a reaction force, equal in magnitude and opposite in direction.”

- each force of the pair (the action force and the reaction force) is applied to a **different body**
- for example, when a jumper generates a force against the ground, the ground generates an equal and opposite force against the jumper, causing the jumper to accelerate upward

$$\begin{aligned}\Sigma F &= N - W \\ &= 687 - 687 = 0 \text{ N} \\ a &= 0.00 \text{ m/s}^2\end{aligned}$$



ground reaction **1.00 × body weight** · net **0 N** · **a = 0.00 m/s²**

The two big arrows are the **action and reaction pair**, and they are always equal and opposite — but they act on **different bodies**, so they never cancel each other. Only the forces on the jumper decide what the jumper does, and there the ground reaction is up against weight. Right now they balance exactly and nothing accelerates.

Body mass ————— **70 kg**

3rd Law: The Law of Reaction



When a jumper generates a force against the ground, the ground generates an equal and opposite force against the jumper.